

Data structure and Algorithms

Module 1 : Overview of Data Structures

1. Stacks,
 2. Queues,
 3. linked lists
 4. doubly linked lists
 5. Applications.
 6. Analysis and Efficiency of algorithms
 7. Asymptotic Notations
 8. Time complexity of an algorithm,
 9. Analysing Recursive Programs using various strategies
 10. Divide and Conquer Paradigm: Divide and conquer recurrence equations and their solutions
 11. Review of various sorting techniques and its time complexity.
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Module 2 : Trees and Graphs

1. Basic terminology& its applications,
2. Binary Trees
3. Binary Search Trees
4. Red-Black Trees
5. AVL,Trees and B Trees,
6. spanning trees: Prim's and Kruskal's algorithm.
7. Terminology

8. representations
 9. traversals
 10. Basic graph algorithms:
 11. Depth first search
 12. Breadth first Search and its analysis,
 13. single source and all-pair shortest path problem.
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Module 3 : Dynamic and Greedy Approach

1. Comparison between Dynamic and Greedy Approach to solve a problem
 2. Matrix Chain Multiplication
 3. Longest Common subsequence and 0/1 Knapsack Problem
 4. Activity Selection Problem
 5. Huffman Codes, and fractional Knapsack Problem.
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Module 4 : E String Matching Algorithms and Approximation Algorithms

1. Naïve
 2. Rabin Karp and Knuth Morris and Pratt 19% paradigm
 3. Vertex-Cover algorithm and Set Covering Problem.
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Module 5 : NP-Completeness

1. Polynomial Time
2. NP-completeness proofs and problems
3. NP-Hard and SAT problems.

